

Introduction and Data

There are several protein targets in drug discovery which can be used as indicators of drug toxicity, and all drugs need to be tested against these targets to prove their safety. One of these targets is the voltage gated inwardly rectifying potassium channel KCNH2 or hERG (human ether α -go-go -related gene). This is the alpha subunit of a potassium ion channel. It controls the repolarization of the current in cardiac action potential and coordinates the heart's beating. If this channels activity is inhibited, it can lead to fatal disorders such as long QT syndrome which can lead to irregular heartbeats, fainting, seizures or death. The hERG channel has a large cavity in the conducting pore which can accommodate various drug like molecules which can inhibit the activity. Residues such as tyrosine and phenylalanine are exposed which can form strong hydrophobic interactions with various drug like molecules. Various drugs have the potential to block this channel and have fatal effects. Therefore, predicting the effect on hERG is an important step in a drug discovery pipeline to filter out toxic drug candidates.

This project aims to develop a pipeline which can identify chemical structures strongly associated with inhibition of hERG through unsupervised machine learning methods and using the information of those chemical structures to train machine learning models to accurately predict the effect on hERG activity. The central question is if knowledge from the chemical space structure from unsupervised machine learning can improve supervised prediction.

Two main datasets have been used for this project: ChEMBL hERG bioactivity dataset (9602 unique compounds with 2164 blocker and 7438 non-blockers) for the unsupervised machine learning and for training and validation and the Therapeutics Data Commons (TDC) dataset (529 compounds) on hERG activity as an external test set to evaluate the model performance and generalization with an unseen data set. This dataset had binary labels for blockers (347 blockers) and non-blockers (182). All the common compounds between TDC and ChEMBL data was removed.

For ChEMBL, the data was filtered for binding assays (Type B). All values were converted to of pIC_{50} values. The standard threshold of $pIC_{50} > 6.0$ was used to divide the data into blocker vs non- blockers.

Methods

Unsupervised Machine Learning

A hyperparameter search was initially conducted to find the best combination of features which show the clearest clustering and separation between toxic and non-toxic clusters. The hyperparameters chosen were ECFP4, Jaccard distance, $n_neighbor = 15$ and $pca = 100$. The fingerprints were computed via RDKit's rdFingerprint Generator with feature-based atom invariants. This produces 2048 binary fingerprints.

To embed the data into a chemical space, incremental PCA (scikit-learn) was applied to project the 2048-dimensional fingerprints to 100 principal components. UMAP (umap-learn) was used to organize those 100 PCs into a 2-dimensional map while maintaining the same grouping.

Clusters were identified through Hierarchical Density Based Spatial Clustering of Applications with Noise (HDBSCAN). A cluster was defined as having at least 30 compounds in it with $min_samples$ (measure of density) = 10.

A cluster was considered toxic if $\geq 60\%$ of the compounds had been designated as blockers.

Point Biserial Correlation was used to evaluate the fingerprints to evaluate if any bits were highly correlated with toxicity. The top 300 bits correlated with toxicity were used in the supervised machine learning models.

To identify specific chemical structures, Murcko Scaffolds were computed (`rdkit.Chem.Scaffolds.MurckoScaffold`). This gives us an idea of the basic skeletal shape of the drug and which structures can be associated with high toxicity. Toxic rate was calculated by dividing the number of “blocker” designated compounds which possess that scaffold by the total number of compounds which have that scaffold. The baseline was taken as 22.5 (22.5% of the drugs from ChEMBL were blockers). Any scaffold with a toxic rate significantly greater than 0.225 could be important. Another factor that was taken into consideration was how many minimum number of compounds need to have that scaffold: at least 5 compounds need to have a scaffold. Based on the `toxic_rate`, enrichment and `pIC` values, all the scaffolds were ranked.

Supervised Machine Learning.

Three Autogluon tabular predictor models were trained to predict binary hERG toxicity labels: Model A: Baseline model which used all 2048 FCFP4 bits as input features.

Model B: UMAP informed model. The 300 FCFP4 bits more enriched in the toxic UMAP clusters were used as an input.

Model C: UMAP informed model with feature reweighting for the top 300 bits from the unsupervised ML parts.

All models used the `best_quality` preset, time limit of 600 seconds. The ChEMBL dataset was split into 80% training and 20% validation with stratified sampling.

Model A (all 2,048 bits) serves as the primary baseline for Model B and C; any improvement in AUROC for Model B or C on TDC would indicate that UMAP-guided feature selection generalizes better than brute-force feature use. Internal ChEMBL validation AUROC vs. external TDC AUROC gap is an indication of the generalizability of the models.

TDC compounds were also projected onto the UMAP created in the unsupervised ML section to see if the blockers occupied the same space. A Mann-Whitney U test was used to test if the blockers and non-blockers of TDC occupy distinct regions on the ChEMBL chemical space map. For every TDC molecule, the identity was decided by the 10 nearest compounds. Continuous scores were computed ranging from 0 to 1. The Mann Whitney U test measured if TDC compounds that are known blockers ($Y=1$) tend to get higher scores than TDC compounds that are known non-blockers ($Y=0$).

Results

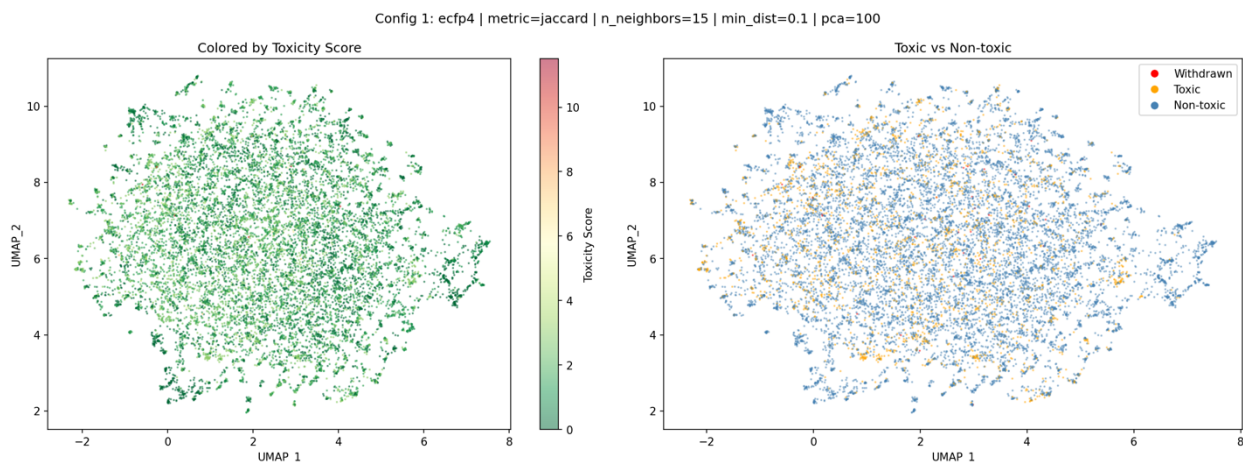


Figure 1 UMAP Projection from Unsupervised ML

The resulting embedding revealed a continuous chemical space with small clusters interspaced. There is no clear separation between toxic and non-toxic groups. HDBSCAN clustering (min_cluster_size=30, min_samples=10) identified 131 stable clusters from this embedding, with 1,964 compounds (20.5%) assigned as noise. 8 (6.1%) were designated as toxic clusters (clusters where > 60% of member compounds are hERG blockers ($pIC_{50} \geq 6.0$)). The remaining 123 clusters had blocker rates below this threshold, with a mean blocker rate of 0.231 across all clusters. The maximum blocker rate observed in any single cluster was 0.885.

Bit	Absolute Correlation
454	0.3249
38	0.294802
315	0.287784
1338	0.259297
88	0.248337
984	0.240168
293	0.237200
450	0.214932
1314	0.213646

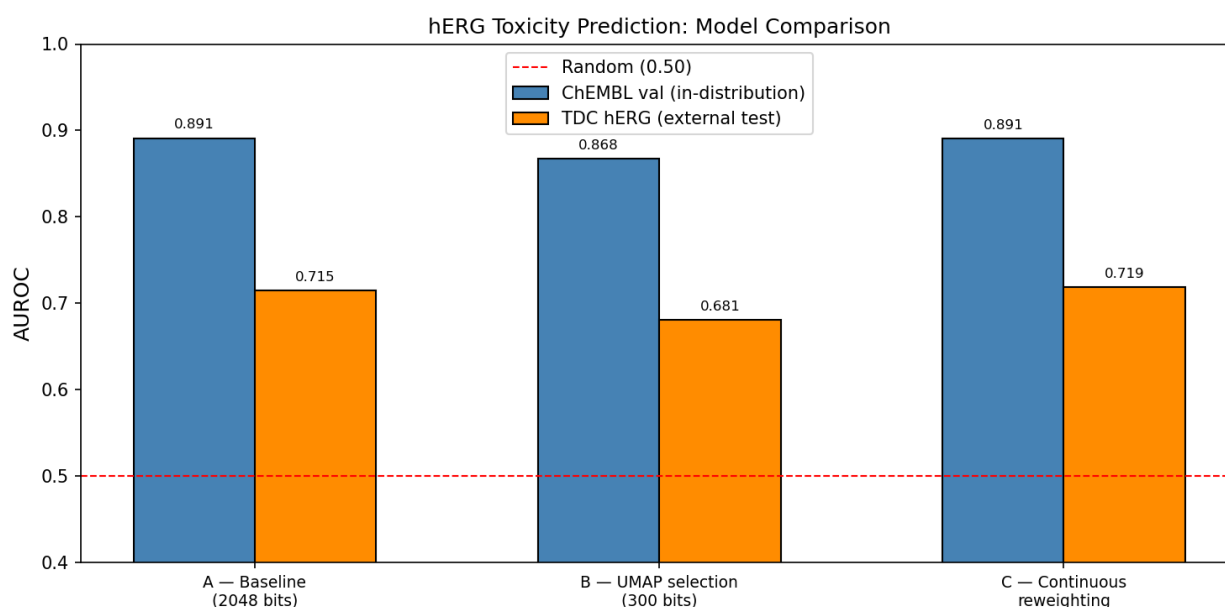
Figure 2 Top 10 bits positively correlated with toxicity

Point-biserial correlation was computed between each of the 2,048 ECFP4 bits and the binary in_toxic_region label derived from HDBSCAN cluster assignments. The top 300 bits by absolute correlation were selected for downstream use in Models B and C. Correlations ranged from 0.035 to 0.325, with a mean of 0.070 across the selected bits. Bit 454 showed the strongest association ($|r| = 0.325$), followed by bits 38, 315, and 1338. Notably, all retained correlations were positive, indicating that the presence of specific substructural features (no absences) is associated with toxicity.

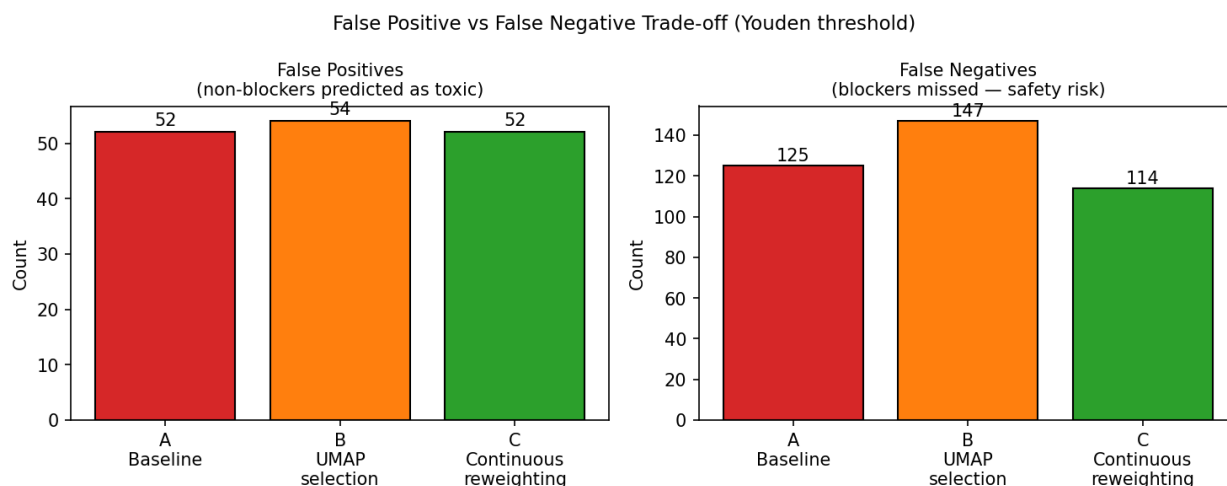
scaffold	count	n_toxic	mean_pIC50	toxic_rate
<chem>c1ccc(OC2CC3CCC(C2)N3Cc2cccs2)cc1</chem>	5	5	8.577061366896130	1.0
<chem>O=c1cc(OCc2ccccc2)ccn1-c1ccc(OCCN2CCCC2)cc1</chem>	8	8	8.509101376014140	1.0
<chem>O=C(COC1ccc(C(=O)c2ccccc2)cc1)Nc1ccccc1</chem>	5	5	7.6305584776919100	1.0
<chem>c1cnc2c(OC3CCNCC3)cccc2c1</chem>	18	17	6.8850010537206500	0.9444444444444440
<chem>c1ccc(-n2cc(Cn3ccnc3)nn2)cc1</chem>	69	64	7.804160510045210	0.927536231884058

Figure 3 Highly Correlated Murcko Scaffolds

Murcko scaffolds were computed to identify the core structures strongly associated with hERG toxicity. Scaffolds represented by at least 5 compounds were ranked by toxic_rate and secondarily by mean_pIC50. Three scaffolds achieved a 100% blocker rate. All these scaffolds have a basic nitrogen capable of interacting with Tyrosine and Aspartate in the channel pore, flanked by hydrophobic aromatic groups that occupy the inner cavity which can interact with tyrosine and phenylalanine.



Model A achieved a validation AUROC of 0.891 and an external TDC AUROC of 0.715. Model B achieved a validation AUROC of 0.868 and TDC AUROC of 0.681. The reduction relative to Model A suggests that the other still contribute meaningful information. Model C has the same baseline validation AUROC of 0.891 and the highest external TDC AUROC of 0.719 which is only marginally higher than Models A and B on the held-out test set.



False positive counts (non-blockers incorrectly predicted as toxic) were similar across models: 52 for Model A, 54 for Model B, and 52 for Model C. False negative counts (true blockers missed by the model, clinically more dangerous) shows more variation: 125 for Model A, 147 for Model B, and 114 for Model C. Model B's increase in false negatives relative to Model A is consistent with its lower TDC test score. Model C achieved the lowest false negative count of 114. Model C's reduction in false negatives along with its marginal TDC AUROC gain might make it the preferred model from a safety perspective, even though its overall discrimination is comparable to Model A.

Discussion:

The main hypothesis was that structural patterns identified without label supervision could improve classifier performance on an external benchmark. The results partially support this hypothesis, with Model C achieving the best external AUROC and the lowest false negative count among the three models tested.

Of 131 HDBSCAN clusters, only 8 were enriched for blockers but those clusters could capture coherent chemical structures and scaffolds which have been known to interact with the hERG channel pore. The Mann-Whitney U test ($p = 0.033$) also showed that TDC blockers projected onto the UMAP matched well with the ChEMBL chemical space. The UMAP geometry captured generalizable structural information. The fingerprint enrichment analysis showed that the top 300 UMAP-informed bits span a correlation range of 0.035 to 0.325. These are not very high values which is consistent with what is seen in experimental data: various structures can interact with hERG and it's likely a combination of various bits and structures.

Comparing the three supervised models; Model B's hard cutoff at 300 bits reduced performance on both the validation set and TDC benchmark relative to Model A, and substantially increased false negatives from 125 to 147. This suggests that the discarded 1,748 bits still collectively contribute meaningful information. Model C addressed this by retaining all bits and emphasizing the contribution of UMAP-informed features through continuous weights proportional to their correlation. The validation performance matched the baseline with marginal improvements in

TDC generalization and a reduction in false negatives to 114. In a safety-critical screening every false negative represents a compound having undetected cardiac risk.

The consistent gap between in-distribution validation AUROC (~ 0.89) and external TDC AUROC (~ 0.71) across all three models could be a sign of overfitting or it could also show a significant difference between the structure of the two datasets. The ChEMBL training data contains 22.5% blockers. The TDC benchmark contains 68.8% blockers. A model trained on the ChEMBL distribution would not be able to translate well on significantly different distributions.